

Foiling data and TracAPI

For the TracTrac's point of view, the Bravo device is just new type of device. We are going to add to our event manager the concept of 'tracker type' that has to be attached to the trackers and it has basically two properties: *name* and *dimensions*. *Name* is just a label used to identify the type of device and *dimensions* is the list of variables that this tracker supports. Example of types:

- *Position*, with the dimensions time, lon, lat, speed and direction
- *Bravo*, with the dimensions time, lon, lat, speed, direction, pitch, roll, ride height....(this list of dimensions has to be defined by SAP)
- *Wind*, with the dimensions time, lon, lat, wind, direction, measure (this is just an example of other type)

By default, the type of the current trackers is *Position*.

At this moment is it possible to attach one of more trackers to both a control and a competitor and using this functionality we can also attach trackers with the type *Bravo*. These trackers also have an identifier and they can be imported in the same way that the *Position* tracker is imported at this moment (e.g: though a CSV import)

The Event Manager has to include in some pages the information of the device type. The following image shows how is going to appear this info in the detail of an entry:

Silverrudder 2016 - Simulation | Event | **Devices** | Races | Viewer ▾ | Details ▾ | Social ▾ | Status | DB

> Edit entry: Mike Bartels Cancel

Course
Silverrudder ▾

Individual start time

Entry status
RAC (Racing) ▾

Status time
2017-09-27

Metadata

Save

Existing tracking sessions

+

Tracking device: **8** (860599000321323)
Sessions starts: 2017-09-27 07:10:48
Sessions ends: 2017-09-30 21:00:00
Priority: 2
Type: **Position**

Tracking device: **127** (860599000092049)
Sessions starts: 2017-09-27 07:10:48
Sessions ends: 2017-09-30 21:00:00
Priority: 1
Type: **Bravo**

TracAPI

The interface *IRaceSubscriber* has to be extended with a method to download the new data stream. We suggest to name this stream "*SensorData*" (it is a generic word that can be used in other contexts):

```
public interface IRaceSubscriber extends ISubscriber {  
  
    public void subscribeSensorData(ISensorDataListener sensorDataListener,  
    SensorDataType sensorDataType);  
  
    public void unsubscribeSensorData(ISensorDataListener sensorDataListener)  
}
```

SensorDataType is an enumerate with the values *Position*, *Bravo*, *Wind*... and in general, with the same list of "device types" supported by the event manager.

ISensorDataListener is the listener used to receive the data stream, which implementation is:

```
public interface ISensorDataListener {  
  
    public void gotSensorData(IRaceCompetitor raceCompetitor, ISensorData  
sensorData);  
  
    public void gotSensorData(IControl control, ISensorData sensorData, int  
controlPointNumber);  
}
```

The initial idea is that the sensor data has to be attached to a control or to a competitor but in a future it can be also extended with new object types (e.g: wind sensor).

ISensorData has getters to retrieve all the supported dimensions:

```
public interface ISensorData {  
  
    public long getTime();  
  
    public IPosition getPosition();  
  
    public float getPitch();  
  
    public float getRoll();  
  
    public float getRideHeight();  
  
    ...  
  
    public float getDimensionN();  
}
```

Thoughts

- We are also considering to add a method *IRaceSubscriber.subscribeSensorData* without the argument *SensorDataType* which meaning is: give me all the sensor data, not matter the device type. With this method, if we add in a future a new device type, any third app would be able to download the stream without update TracAPI if the dimension is already defined in the interface *ISensorData* (to solve this issue we can also add the method "public Object getDimensionAt(int index)" to the *ISensorData* interface).